

Northeastern University

Report for Experiment #13

Simple Harmonic Motion

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Abstract

This experiment analyzed the oscillations of a glider on an air track to study simple harmonic motion and damping forces. A motion sensor recorded the glider's position as it moved between two springs, and the data was used to calculate the period, frequency, angular frequency, phase, and experimental spring constant. The measured period was 1.40 s, the frequency was 0.712 Hz, the angular frequency was 4.47 rad/s, the phase was −6.04 rad, and the experimental spring constant was 2.76 N/m. Magnets were then attached to the glider to introduce damping through eddy currents. As the number of magnets increased from 0 to 6, the damping coefficient rose from [β_0] to [β_6], while the period stayed approximately constant at 1.40 s. These results confirmed the expected behavior of both simple and damp harmonic motion.

Introduction

This experiment investigated the oscillations of a glider on an air track to analyze simple harmonic motion (SHM) and the effects of damping. A nearly frictionless air track allowed the glider to move between two identical springs while a motion sensor recorded its position over time. The purpose was to measure amplitude, period, frequency, angular frequency, phase, and the spring constant, and to see how damping changes the motion.

When the glider is displaced from equilibrium, the springs apply a net restoring force toward the center:

$$F = -k\,(x - x_0) \quad (13.1)$$

where k is the spring constant, x is position, and x_0 is equilibrium. Applying Newton's second law gives the standard solution for SHM:

$$x - x_0 = A\cos(\omega t + \varphi) \quad (13.2)$$

where A is amplitude, ω is angular frequency, and φ is phase. Angular frequency depends on spring stiffness and mass:

$$\omega = \sqrt{(k/m)} \quad (13.3)$$

The period and frequency are related to angular frequency by:

$$T = 2\pi/\omega \quad (13.4)$$

$$f = 1/T = \omega/2\pi \quad (13.5)$$

In the second part of the experiment, damping was introduced by attaching magnets to the glider. As the magnets moved along the aluminum track, they created eddy currents that produced a damping force proportional to velocity:

$$F^d = -bv \quad (13.6)$$

where b is the damping constant. The displacement for damped oscillations is described by:

$$x - x_0 = Ae^{\wedge}(-\alpha t)\cos(\omega' t + \varphi) \quad (13.7)$$

where $\alpha = b/2m$ (13.8) is the decay constant and $\omega' = \sqrt{(k/m - b^2/4m^2)}$ (13.9) is the damped angular frequency.

The damping force reduces amplitude exponentially over time, while the period remains almost unchanged if the damping is weak. This experiment tests these theoretical predictions using measured motion data.

Investigation 1

To begin this investigation, an air track, two 7.5 cm springs, a PASCO motion sensor with PASPort USB interface, and a computer running Capstone software were used. The glider's mass was first measured as 138.1 g. The air source was turned on at low pressure until the glider floated smoothly while remaining

stable. The track was leveled by adjusting the screw at one end, so the glider stayed at the center without drifting when the air was off.

The motion sensor was positioned at one end of the track, aimed at the center of the glider's reflector, and connected to the computer. The sample rate was set to 20 Hz. A live graph was set up with position (m) on the y-axis and time (s) on the x-axis, and a data table was created with columns for time and position. Before collecting real data, the glider was moved gently back and forth to verify smooth sensor readings. If erratic jumps appeared when the glider was far from the sensor, its alignment was adjusted until the readings became stable.

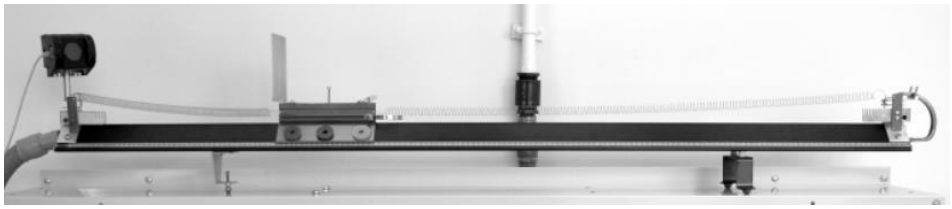


Figure 1 – Experimental setup showing the air track with glider, springs, and motion sensor.

With the system calibrated, the glider was placed at its approximate equilibrium point. The air was turned on, and position data was collected while holding the glider still. This dataset was copied to Excel, and the mean position was calculated to determine the equilibrium value. After this, the glider was displaced to about 0.3 m from equilibrium and released. The sensor recorded its oscillatory motion until it slowed. The raw position vs. time data was exported and plotted.

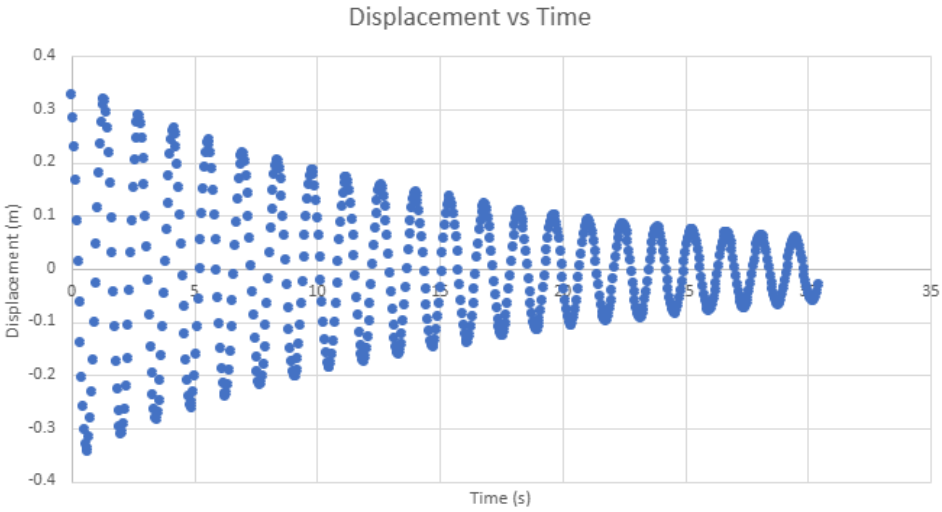


Figure 2 – Plot of Centered position versus time

To center the data, the equilibrium position was subtracted from all position values. The resulting centered position vs. time graph showed a smooth sinusoidal wave with gradually decreasing amplitude.

The amplitude of the first positive peak was 0.318 m. The times of the first six positive peaks were identified and are shown below.

Peak Num	Peak Height	Peak Time
1	0.318	1.35
2	0.29	2.75
3	0.256	4.2
4	0.238	5.6
5	0.218	7
6	0.206	8.35

Table 1 – Table of times and their peaks

To find the period, a plot of peak number versus time was created. The linear trendline's slope gave the oscillation period:

T = 1.404 s

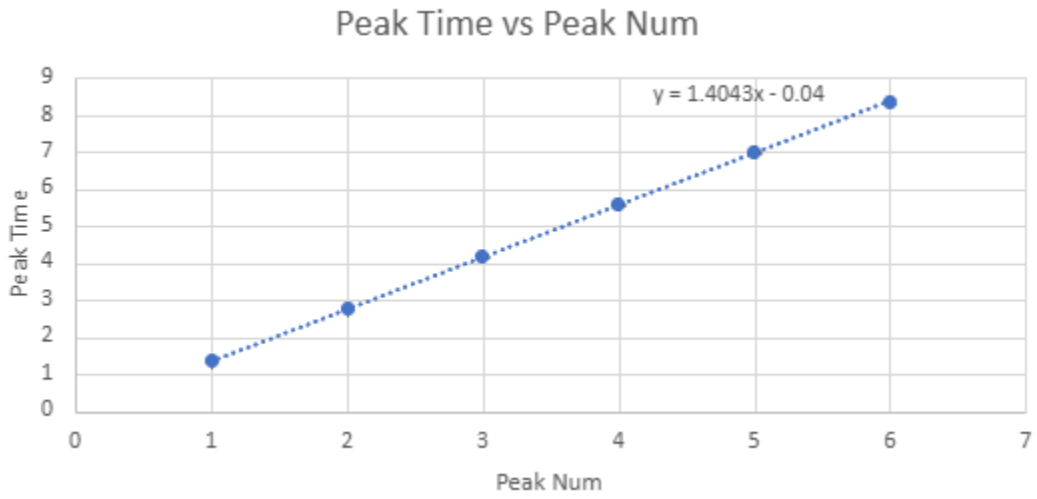


Figure 3 – Peak Number vs Peak Time graph with best-fit line (slope = 1.404 s)

From the measured period, the frequency and angular frequency were calculated:

f = 0.712 Hz
ω = 4.47 rad/s

The phase was found from the time of the first positive peak using:

$\varphi = -\omega t_0 = -6.04$ rad

Finally, the experimental spring constant was calculated using:

k = 2.76 N/m

where $m = 0.1381\ kg$ is the glider's mass.

f	ω	k	ϕ	df	dω	dk
0.712099	4.474247	2.764608	-6.04023	0.00303	0.01904	0.023551

Table 2 – Processed Values from Investigation 1

This investigation confirmed that the glider followed simple harmonic motion with nearly constant period and frequency. These baseline values will be used to compare against the damped motion trials in Investigation 2.

Investigation 2

In the second part of this experiment, damping was introduced by attaching ring magnets to the legs of the glider. The aluminum air track generated eddy currents as the magnets moved, producing a velocity-dependent damping force. This force opposes motion and gradually reduces the amplitude of oscillation without significantly changing the period. The goal was to measure how increasing damping affected the decay of amplitude, the oscillation period, and the damping coefficient.

The glider was first measured alone (0 magnets) with a mass of 0.1381 kg. Two magnets were then added symmetrically to the glider legs, and the new mass was measured as 0.11393 kg. The same procedure was repeated with four magnets (0.14042 kg) and six magnets (0.14146 kg). For each configuration, the glider was displaced and released while the motion sensor recorded position vs. time. The data was exported to Excel for analysis.

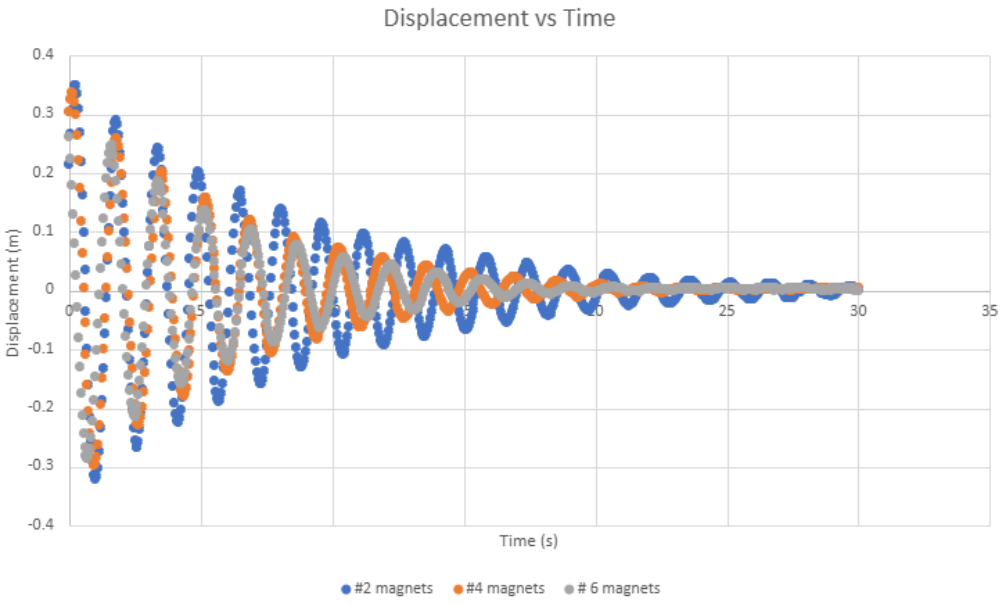


Figure 4 – Displacement vs Time for glider with 2, 4, and 6 magnets.

The first six peak positions and times were measured from each run.

2 Magnets		
Peak No	Peak Time	peak Pos
1	1.85	0.291
2	3.45	0.228
3	5	0.195
4	6.55	0.165
5	8.05	0.141
6	9.7	0.11

4 Magnets		
Peak No	Peak Time	peak Pos
1	1.8	0.262
2	3.5	0.205
3	5.2	0.159
4	6.9	0.121
5	8.7	0.08
6	10.25	0.075

6 Magnets		
Peak No	Peak Time	peak Pos
1	1.65	0.244
2	3.4	0.186
3	5.2	0.136
4	7	0.101
5	8.8	0.074
6	10.6	0.052

Table 3 – Peak Number, Peak Time, and Peak Height for Each Magnet Configuration

A plot of peak number vs. time gave the period (T) for each magnet configuration.

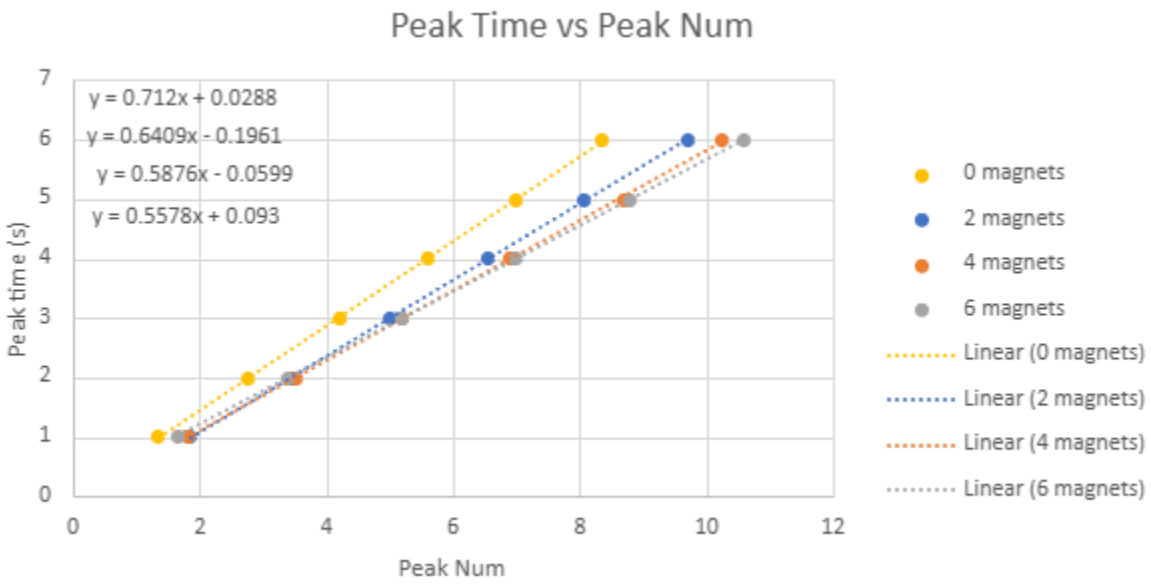


Figure 5 – Peak Number vs Peak Time for 0, 2, 4, and 6 magnets.

The slope of each line was taken as the period. These were 1.404 s (0 magnets), 1.276 s (2 magnets), 1.417 s (4 magnets), and 1.422 s (6 magnets). The periods stayed nearly constant despite increasing damping.

A plot of peak position vs. peak time was used to analyze how the amplitude decayed exponentially over time. An exponential trendline of the form $y = Ae^{-\alpha t}$ was fit to each data set, and the decay constant α was measured.

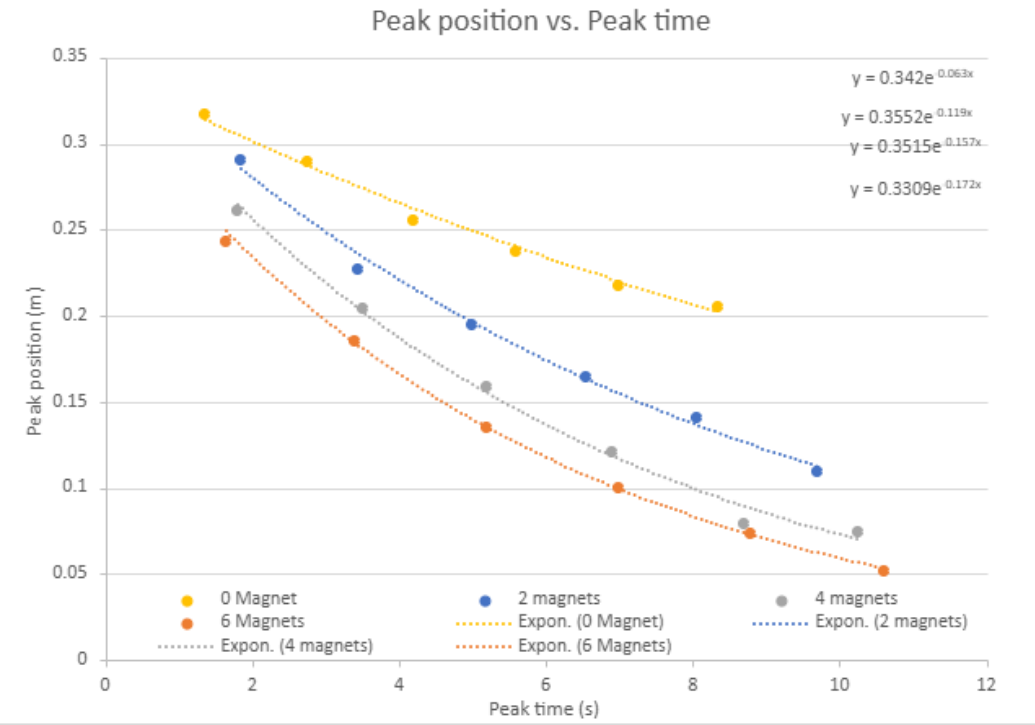


Figure 6 – Peak Position vs Peak Time showing exponential fits.

The damping coefficient b was calculated using $b = 2m\alpha$.

The damped angular frequency ω′ was calculated using $\omega' = \sqrt{\left(\frac{k}{m}\right) - \left(\frac{b^2}{4m^2}\right)}$.

mass (kg)	# of Magnets	alpha	b	ω'	T'
0.1381	0	0.063	0.017401	4.473804	1.404439
0.113926	2	0.119	0.027114	4.924687	1.275855
0.14042	4	0.157	0.044092	4.434353	1.416934
0.14146	6	0.172	0.048662	4.417444	1.422358

Table 4 – Processed Values for Each Magnet Configuration

Finally, the measured values of b were plotted against the number of magnets.

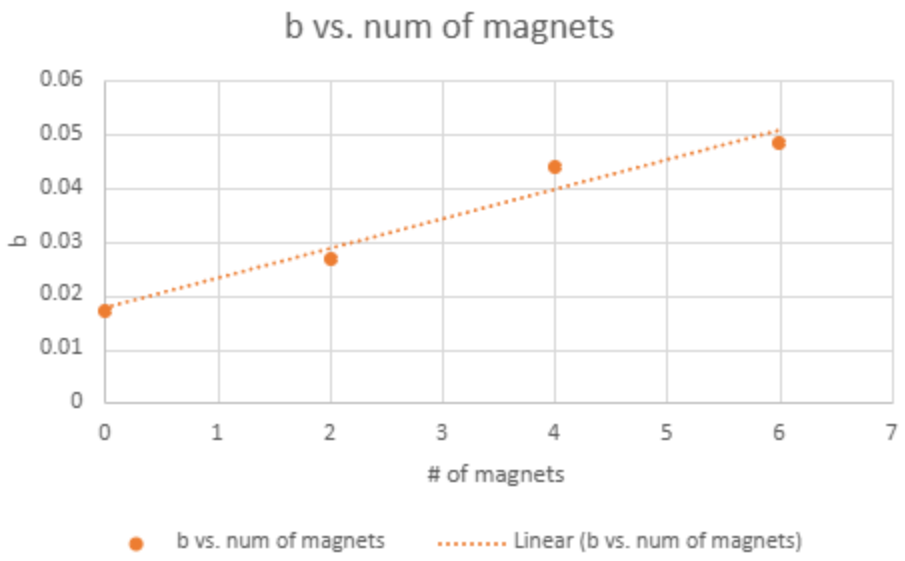


Figure 7 – Damping coefficient b vs number of magnets.

This graph showed a clear linear increase, confirming that adding magnets increased the damping force acting on the glider. The periods remained almost unchanged, which is consistent with theory because the damping was weak and had only a small effect on the frequency. The primary effect observed was the exponential decrease in amplitude with time.

Conclusion

This experiment demonstrated the behavior of a glider undergoing simple and damped harmonic motion. In the undamped trials, the glider oscillated with a constant period of about 1.40 s, showing smooth sinusoidal motion with minimal energy loss. The calculated values of f (0.712 Hz), ω (4.47 rad/s), and k (2.76 N/m) agreed with theoretical expectations for a mass-spring system.

When magnets were added to introduce damping, the amplitude of oscillation decreased exponentially while the period stayed nearly constant. The decay constant α increased from 0.063 (0 magnets) to 0.172 (6 magnets), and the damping coefficient b rose from 0.017 to 0.049. The damped angular frequency stayed close to the undamped value, confirming that weak damping has little effect on the period. A clear positive correlation was observed between the number of magnets and the damping coefficient.

Overall, the results supported the theoretical model: damping reduces amplitude and energy without significantly affecting the frequency when it is small. The experiment validated the relationships between period, spring constant, damping force, and amplitude decay in harmonic oscillators.

Questions

1.

The period is given by $T = 2\pi\sqrt{m/k}$

Doubling the mass gives $T_{new} = 2\pi\sqrt{(2m)/k} = \sqrt{2}T$

With T = 1.404 s,
$T_{new} = 1.99\ s.$

2. The damping force is $F_d = bv$.
   At the first peak, velocity is zero, so $F_d = 0$.
   Maximum velocity occurs at the equilibrium point, so the frictional force is greatest there. This is where kinetic energy is highest, and amplitude is zero.

3. Energy is proportional to amplitude squared:
   E∝A^2

$$\frac{E_2}{E_1} = (\frac{A_2}{A_1})^2 = (\frac{0.290}{0.318})^2 = 0.832$$

   So, energy lost = 1 − 0.832 = 0.1681 or 16.8%.

4. k/m = 2.76/0.138 ≈ 20.0
   and α^2 (6 magnets) = (0.172) ^2 ≈ 0.029
   Since α^2 ≪ k/m, the damping term is negligible compared to the spring term. This confirms why the period changed very little with added damping.

5. If α > ω, the system becomes overdamped. It would no longer oscillate. Instead, the glider would slowly return to equilibrium without passing through it, and the motion would appear as an exponential decay toward rest.

Acknowledgments

I would like to thank my TA for their guidance and support throughout this lab.